

Cedrick Tchakonte Tchiegue

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EDUCATION

ENSTA - Paris-Saclay Campus, Palaiseau, France

Aug 2024 - Aug 2025

- Second-year engineering student in Artificial Intelligence and Cyber-Physical Systems
- **Coursework:** Machine Learning, Image Recognition, Microprocessor Architecture, Statistical Learning, Control Theory, Robotic Navigation.

National Advanced School of Engineering, Yaoundé, Cameroon

Oct 2020 - Aug 2024

- Intensive preparatory program in mathematics and physical sciences for international competitive entrance exams
- Admissible at the École Polytechnique entrance exam; admitted to IP Paris (ENSTA Paris, Télécom Paris, ENSAE)
- **Coursework:** Linear Algebra, Data analysis, Mechanics, Differential Equations, Probability and Statistics..

EXPERIENCE



Machine Learning Research Intern, Stellantis, GrEEn-Campus Poissy, France

Feb 2026 - present

- Built a multimodal dataset from 3D scans, meshes and impact test data (Euro NCAP / China NCAP / Japan NCAP frameworks).
- Applied transfer learning (ResNet, ShuffleNet, EfficientNet) and designed 3D CNN/GNN surrogate models to predict pedestrian protection metrics.
- Benchmarked against physical tests and finite element simulations to support early-stage safety optimization.



Junior AI Engineer, TAEF (Junior Enterprise of ENSTA), Palaiseau, France

Nov 2025 - present

- Developed a Graph RAG system for European financial regulation, building knowledge graphs from regulatory corpora to model relationships between directives, articles and legal entities.
- Engineered LLM-based entity and relation extraction pipelines to populate graph structures from unstructured legal texts.
- Combined vector search with graph traversal for hybrid retrieval, improving relevance over standard RAG.



Junior AI Research Engineer, Objectware, Paris, France

Sept 2025 - Jan 2026

- Created AI-based decision support systems using LLMs and NLP for predictive analytics and enterprise automation.
- Fine-tuned small language models using LoRA/QLoRA for domain-specific tasks with reduced compute cost.



R&D Intern - AI for Embedded Systems, STMicroelectronics, France

May 2025 - Aug 2025

- Built AI-assisted workflows using LLMs, RAG and vector databases to automate documentation analysis and model generation for C++ Digital Twin simulations.
- Integrated the solution into SOC Digital Twin development across multiple ST divisions (automotive, RF, MCUs).

PROJECTS

Retinal Vessel Segmentation in SLO Ophthalmoscopy, ENSTA

May 2025 - Jun 2025

- Developed automated retinal vessel segmentation using gradient operators and vascular filters for SLO images, validated against IOSTAR ground truth.
- Applied morphological filtering and image processing techniques for precise vascular structure extraction.

Adversarial Attacks on Segmentation Models, ENSTA

Jan 2025 - Feb 2025

- Implemented FGSM-based targeted and untargeted adversarial attacks on DeepLabV3/FCN segmentation models (MS-COCO, PyTorch).
- Evaluated cross-architecture transferability and perturbation impact on segmentation robustness.

4D GPS Navigation System for VTOL Aircraft, ENSTA

Oct 2024 - May 2025

- Designed trajectory optimization algorithms for VTOL aircraft with Technoplane, integrating 3D mapping and dynamic weather modeling.

CERTIFICATIONS

🌟 Machine Learning Specialization, Stanford University & DeepLearning.AI

Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning & Recommender Systems.

🌟 Deep Learning Specialization, DeepLearning.AI

Improving Deep Neural Networks, Structuring ML Projects, Convolutional Neural Networks, Sequence Models.

🌟 **Project Management Certificate (MOOC)**

Strategic planning, execution, monitoring, and risk management methodologies.

TECHNICAL SKILLS

- 📄 **Programming:** Python, C/C++, R, Bash, MATLAB, Java
- ⚙️ **AI & ML:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Hugging Face, Transformers, LangChain
- 👁️ **Computer Vision:** OpenCV, image segmentation, feature detection, morphological filtering
- 💬 **NLP & LLMs:** RAG, Graph RAG, vector databases, knowledge graphs, prompt engineering
- 🔧 **Tools:** Git, Docker, Linux, LaTeX, SQL, Neo4j
- 🗣️ **Languages:** French (Native), English (C1 - Advanced), German (A2 - Elementary)

EXTRACURRICULAR ACTIVITIES AND INTERESTS

- Finance Officer, ENSTAFRIK Association, ENSTA** **Apr 2025 - present**
Management of financial operations, preparation of budgets and coordination of cultural events.
- Volunteer, Ryz'Ôm Joëlettes (Disability Inclusion Association)** **Nov 2024 - present**
Promote disability inclusion through Joëlette wheelchair events and lead awareness training sessions in high schools.
- Personal interests:** Volleyball, cycling, football, chess.